

Divergent dual mass forces a fixed random-section radius

Solution packet for arXiv:2109.14504

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Outcome. The zero-decay half of the source conjecture holds in the stronger proposed generality: if $1 < p \leq \infty$, $q = p^*$, and the nonincreasing semiaxes satisfy $\sigma \notin \ell_q$, then for every fixed codimension n a sufficiently long Gaussian section has radius at least $\sigma_1/2^{1/p}$ with arbitrarily high probability. For $\sigma_j = j^{-\lambda}$ this is exactly $\lambda \leq 1/p^*$, closing both cases left open in the source.

Source question

For $1 \leq p \leq \infty$ and nonincreasing positive σ , put

$$\mathcal{E}_{p,\sigma}^m = \left\{ x \in \mathbb{R}^m : \sum_{j=1}^m |x_j/\sigma_j|^p \leq 1 \right\},$$

with the usual maximum convention for $p = \infty$. If $G_{n,m}$ is an $n \times m$ standard Gaussian matrix, the source asks for the polynomial decay of $\text{rad}(\mathcal{E}_{p,\sigma}^m, \ker G_{n,m})$ uniformly over $m > n$.

The source proves its threshold conjecture except when

$$1 < p < 2, \quad \lambda = 1/p^*, \quad \text{or} \quad p > 2, \quad 1/2 < \lambda \leq 1/p^*.$$

It further proposes zero decay whenever $\sigma \notin \ell_{p^*}$. The exact source passage is reproduced below.

Conjecture. Let $1 \leq p \leq \infty$ and $\sigma_j = j^{-\lambda}$, $j \in \mathbb{N}$, with $\lambda > \frac{1}{s} = (\frac{1}{2} - \frac{1}{p})_+$. Then,

$$\text{decay}(\text{rad}(\mathcal{E}_{p,\sigma}^m, G_{n,m})) = \begin{cases} \text{decay}(\text{rad}(\mathcal{E}_{p,\sigma}^m, n)) & : \lambda > \frac{1}{p^*}, \\ 0 & : \lambda \leq \frac{1}{p^*}. \end{cases}$$

By the discussion prior to the conjecture, it is verified except for the two cases

1. $1 < p < 2$ and $\lambda = \frac{1}{p^*}$,
2. $p > 2$ and $\frac{1}{2} < \lambda \leq \frac{1}{p^*}$.

As a matter of fact, it seems reasonable to conjecture that $\text{decay}(\text{rad}(\mathcal{E}_{p,\sigma}^m, G_{n,m})) = \text{decay}(\text{rad}(\mathcal{E}_{p,\sigma}^m, n))$ as long as $\|(\sigma_j)_{j \in \mathbb{N}}\|_{p^*} < \infty$, while $\text{decay}(\text{rad}(\mathcal{E}_{p,\sigma}^m, G_{n,m})) = 0$ whenever $\|(\sigma_j)_{j \in \mathbb{N}}\|_{p^*} = \infty$. We leave this as an open problem for future investigation.

Organization of the paper

A uniform Gaussian small-ball lemma

Lemma 1. *For every $q > 0$ there are constants $c_q, c, C > 0$ such that, whenever $L \geq Cn$ and $g_1, \dots, g_L$ are independent standard Gaussian vectors in $\mathbb{R}^n$,*

$$\Pr \left\{ \inf_{v \in S^{n-1}} \left(\frac{1}{L} \sum_{j=1}^L |\langle g_j, v \rangle|^q \right)^{1/q} \geq c_q \right\} \geq 1 - 2e^{-cL}.$$

Proof. Let $a = \Pr\{|Z| \geq 1\} > 0$ for a standard Gaussian Z . Choose a fixed $\eta > 0$ so small that $36\eta^2 \leq a/4$, and let $\mathcal{N}$ be an η -net of S^{n-1} with $|\mathcal{N}| \leq (3/\eta)^n$. For each $u \in \mathcal{N}$, a binomial Chernoff bound gives

$$\#\{j : |\langle g_j, u \rangle| \geq 1\} \geq aL/2$$

except on an event of probability $e^{-c_0 L}$. A union bound makes this simultaneous over $\mathcal{N}$ once $L \geq Cn$.

Write $H : \mathbb{R}^n \rightarrow \mathbb{R}^L$, $Hv = (\langle g_j, v \rangle)_j$. The standard Gaussian operator-norm estimate gives $\|H\|_{2 \rightarrow 2} \leq 3\sqrt{L}$ outside an event of probability $e^{-c_1 L}$. Given $v \in S^{n-1}$, choose $u \in \mathcal{N}$ with $\|v - u\|_2 \leq \eta$. Among the at least $aL/2$ indices for which $|\langle g_j, u \rangle| \geq 1$, at most

$$4\|H(v - u)\|_2^2 \leq 36L\eta^2 \leq aL/4$$

can satisfy $|\langle g_j, v - u \rangle| > 1/2$. Thus at least $aL/4$ coordinates obey $|\langle g_j, v \rangle| \geq 1/2$. Hence the displayed empirical q -moment is at least $(a/4)^{1/q}/2$, uniformly in v . $\square$

Main theorem

Theorem 1. *Let $1 < p \leq \infty$, let $q = p^*$, and let $\sigma_1 \geq \sigma_2 \geq \dots > 0$ with $\sum_{j \geq 1} \sigma_j^q = \infty$. For every $n \in \mathbb{N}$ and every $\varepsilon \in (0, 1)$ there is $m > n$ such that*

$$\Pr \left\{ \text{rad}(\mathcal{E}_{p, \sigma}^m, \ker G_{n, m}) \geq \frac{\sigma_1}{2^{1/p}} \right\} \geq 1 - \varepsilon,$$

where $2^{1/\infty} = 1$.

Proof. Write the columns of $G_{n, m}$ as $g_1, \dots, g_m$. Consider the symmetric convex body

$$K_m = \left\{ \sum_{j=2}^m \sigma_j u_j g_j : \|(u_j)_{j=2}^m\|_p \leq 1 \right\} \subset \mathbb{R}^n.$$

Its support function is, with $q = p^*$,

$$h_{K_m}(v) = \left(\sum_{j=2}^m \sigma_j^q |\langle g_j, v \rangle|^q \right)^{1/q};$$

this formula also covers $p = \infty, q = 1$.

Use dyadic blocks $I_k = \{2^k + 1, \dots, 2^{k+1}\}$, of length 2^k . Choose k_0 large enough that Lemma 1 applies on every I_k , $k \geq k_0$, and that the sum of their failure probabilities is at most $\varepsilon/2$. Simultaneously on all blocks $I_{k_0}, \dots, I_K$, monotonicity of σ gives, for every $v \in S^{n-1}$,

$$h_{K_m}(v)^q \geq c_q^q \sum_{k=k_0}^K 2^k \sigma_{2^{k+1}}^q, \quad m = 2^{K+1}.$$

By Cauchy's condensation theorem, the sum on the right tends to infinity with K , because (σ_j^q) is nonincreasing and has divergent sum.

Choose B so that $\Pr\{\|g_1\|_2 \leq B\} \geq 1 - \varepsilon/2$, and then choose K so large that the preceding lower bound for h_{K_m} is at least $\sigma_1 B$. Support-function domination yields

$$K_m \supseteq \sigma_1 B B_2^n.$$

On the intersection of the two good events, $-\sigma_1 g_1 \in K_m$. Consequently there is a tail vector $(u_j)_{j=2}^m$ with $\|(u_j)_{j=2}^m\|_p \leq 1$ and

$$\sum_{j=2}^m \sigma_j u_j g_j = -\sigma_1 g_1.$$

Set $u_1 = 1$ and $x = 2^{-1/p}(\sigma_j u_j)_{j=1}^m$ (with no rescaling when $p = \infty$). Then $x \in \mathcal{E}_{p,\sigma}^m$, $G_{n,m}x = 0$, and

$$\|x\|_2 \geq |x_1| = \sigma_1/2^{1/p}.$$

The two good events have joint probability at least $1 - \varepsilon$. □

Consequence for the source conjecture

If $\sigma_j = j^{-\lambda}$, then $\sum_j \sigma_j^{p^*} = \infty$ exactly when $\lambda \leq 1/p^*$. Theorem 1 therefore supplies a fixed positive lower bound for a suitable m at every n , with arbitrarily high probability.

This forces the source's uniform polynomial decay exponent to be zero. Indeed, a positive exponent ρ would give constants $C, C_1, C_2 > 0$ such that, for every $m > n$,

$$\Pr\{\text{rad}(\mathcal{E}_{p,\sigma}^m, \ker G_{n,m}) \leq Cn^{-\rho}\} \geq 1 - C_1 e^{-C_2 n}.$$

For large n , $Cn^{-\rho} < \sigma_1/2^{1/p}$ and the probability on the right exceeds $1/2$. Theorem 1, with (say) $\varepsilon = 1/4$, gives an m for which the incompatible lower event has probability at least $3/4$.

Thus the two missing regimes listed by the source both have decay zero. Combined with the cases already proved there, this establishes the stated polynomial threshold conjecture for every $1 \leq p \leq \infty$.

Audit notes

- The argument uses the quantifier “for every n , some sufficiently large m ,” exactly as required to disprove any upper estimate uniform in m .
- The first column is separated before the tail body is formed; no circular dependence enters the absorption step.
- Support-function domination has the correct direction: $h_{K_m} \geq R$ on the sphere is equivalent to $K_m \supseteq RB_2^n$.
- At the critical line, each dyadic block contributes constant dual mass; arbitrarily many independent blocks supply the missing logarithmic accumulation.
- No assertion is made for $p = 1$ in the general ℓ_{p^*} formulation. The polynomial conjecture at $p = 1$ was already outside the missing list in the source.

Source

A. Hinrichs, J. Prochno, and M. Sonleitner, *Random sections of ℓ_p -ellipsoids, optimal recovery and Gelfand numbers of diagonal operators*, arXiv:2109.14504v1 (2021), later Journal of Approximation Theory 289 (2023), 105919.